Retinal Illuminance and Visual Acuity: A Literature-Grounded Basis for a Brightness Metric in Pinhole Aperture Design

Optics/Vision Literature Review

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Abstract

A wave-optics simulation of pinhole eyewear for myopia currently uses a reverse-engineered rule that reducing retinal illuminance to roughly 6–9% of full natural-pupil brightness marks the threshold below which visual acuity begins to degrade. This report reviews the peer-reviewed vision-science and clinical literature to test that rule. We find that the literature does not support a percentage-of-baseline acuity threshold; the citable thresholds are *absolute* (in trolands or cd/m^2). High-contrast foveal acuity is nearly flat across the photopic plateau (slope ≈ 0.06 logMAR per decade of luminance) and degrades only as absolute luminance approaches ~ 1 cd/m^2 or the photopic/mesopic boundary. We recommend replacing the 6–9% rule with an absolute retinal-illuminance floor of ~ 300 photopic trolands (equivalently keeping adaptation above the $\sim$3–5 cd/m^2 photopic floor), corrected for pupil dilation and neural brightness adaptation.

1 Introduction: the engineering question

A pinhole aperture placed before the eye improves the retinal image by reducing geometric blur, but simultaneously reduces retinal illuminance because the aperture is smaller than the pupil. The design question is: *what minimum retinal illuminance, expressed as a fraction of the full natural-pupil value under daylight assumptions, must the aperture preserve so that acuity is not degraded?* The simulation currently answers "about 6–9%," a figure tuned to a target range rather than cited. We ask whether the literature supports a percentage threshold at all, and if not, what the closest defensible proxy is.

Retinal illuminance E (trolands) relates to scene luminance L and pupil area A_p by

$$E \;=\; L\,A_p, \qquad [\mathrm{Td}] = [\mathrm{cd/m^2}] \cdot [\mathrm{mm^2}], \tag{1}$$

after Thibos et al. (14). For a pinhole the effective area is $A_{\mathrm{eff}} = \min(A_{\mathrm{aperture}}, A_p)$.

2 Literature review

2.1 Acuity versus illuminance: the classical curve

Shlaer (12) established that foveal acuity rises with $\log I$ and then saturates: at the highest illuminations acuity "remains constant regardless of the changes in $\log I$." The curve is a two-branch (rod, then cone) sigmoid with a rod/cone break near decimal acuity 0.16. The cone branch has a linear range of about three log units and asymptotes near ~ 300 cd/m^2 in later restatements

(9; Webvision). Free-viewing acuity is nearly identical for the three highest retinal illuminances tested, $+2.73, +1.73, +0.73$ log photopic Td.

2.2 The photopic acuity–luminance slope

Several studies quantify how little acuity changes on the plateau:

- Sheedy et al. (11): over 40–600 $\mathrm{cd/m^2}$, log(MAR) is linear in $\log L$; a doubling of luminance (0.3 log units) improves acuity by one letter (0.02 logMAR), i.e. a slope of ≈ 0.067 logMAR per decade.
- Tidbury et al. (15): a one-log-unit change in illuminance produces 0.060 logMAR ($p < 0.001$), measured over 50–8000 lux; the effect grows with uncorrected refractive error (to 0.12 logMAR/decade at +1.5 D).
- Rabin (8): high-contrast acuity stays clinically 20/20 from 100 down to 1 $\mathrm{cd/m^2}$; only below $\sim$1 $\mathrm{cd/m^2}$ does it fall. Small-letter contrast sensitivity, by contrast, falls 8–17$\times$ over the same range.
- Johnson & Casson (5): over 75.0–0.075 $\mathrm{cd/m^2}$ acuity decreases linearly with reductions in luminance; luminance, contrast and blur effects are additive.

The consensus photopic slope is 0.03–0.15 logMAR per decade, centered near 0.06.

2.3 The Rose–De Vries law and its range of validity

For a target of luminance increment ΔL against background L, quantum-noise limiting predicts a constant signal-to-noise ratio when

$$\frac{\Delta L}{\sqrt{L}} = \text{const} \implies C_{\text{th}} \equiv \frac{\Delta L}{L} \propto \frac{1}{\sqrt{L}}, \tag{2}$$

the De Vries–Rose law (3; 10), a slope of $-\frac{1}{2}$ in $\log C_{\text{th}}$ vs. $\log L$. Van Nes & Bouman (16) measured the transition to Weber's law ($C_{\text{th}} = \text{const}$, slope 0) at $\sim$300 Td: "For mean retinal illuminances B_0 smaller than 300 td, threshold modulation M at a given frequency is found to increase in proportion to $B_0^{-1/2}$ (de Vries–Rose law). For B_0 greater than 300 td M remains a constant fraction of it (Weber–Fechner law)." Donner & Fagerholm (4) argue the $\sqrt{\cdot}$ behavior reflects retinal gain control more than pure photon noise. **Consequence:** at genuine daylight (Weber regime) acuity and contrast are largely independent of light level, so a shot-noise brightness argument for acuity loss does not apply until illuminance falls into the mesopic De Vries–Rose range.

2.4 Pinhole and small-aperture studies: brightness vs. blur

Kim et al. (6) measured 48 eyes with 0.9-mm-aperture pinhole glasses: uncorrected distance acuity improved (0.44 $\rightarrow$ 0.19 logMAR) from blur reduction, while the dimming reduced contrast sensitivity by 11.6–34.6% (3–18 cpd), degraded visual-field mean deviation ($-0.48 \rightarrow -4.22$ dB) and stereopsis, with the pupil dilating from 3.6 to 6.0 mm. This directly separates the acuity-improving (blur) and performance-degrading (brightness) effects. Artal & Manzanera (2) and Manzanera et al. (7) showed perceived brightness through a 1.6-mm aperture is 38.5–46.9% (vs. a geometric 28.4%), a neural-adaptation gain of $\times$1.24–1.51: the *functional* dimming is smaller than the geometric ratio.

2.5 Fractional reductions: ND filters and tints

von Noorden & Burian (17) showed an ND filter lowers normal-eye acuity by a roughly constant amount. A modern replication found a 3.0 ND filter ($\approx 0.1\%$ transmission) significantly reduced acuity in all eyes, whereas a 0.4 ND filter ($\approx 40\%$ transmission) did not. General-purpose sunglasses (ANSI Z80.3, 8–40% transmittance) leave photopic acuity intact. No study defines a percentage-of-baseline acuity threshold.

2.6 Contrast sensitivity vs. luminance

Van Nes & Bouman (16) found CSF peak sensitivity saturating near 900 Td and the high-frequency (acuity) cutoff rising up to ~ 300 Td. Wuerger et al. (19) found achromatic contrast sensitivity increasing with luminance up to $\sim 200\,\mathrm{cd/m^2}$ then declining, with the CSF peak shifting to higher frequency and amplitude as luminance rises. Contrast sensitivity is a more sensitive early marker of dimming than high-contrast acuity.

2.7 Photopic/mesopic/scotopic boundaries

CIE 191:2010 and IES TM-12-12 define the mesopic range as 0.005–5.0 $\mathrm{cd/m^2}$; photopic above $\sim 5\,\mathrm{cd/m^2}$, scotopic below $\sim 0.005\,\mathrm{cd/m^2}$. Rods saturate between ~ 3 and 120–300 $\mathrm{cd/m^2}$ (1; 13).

3 Evaluation of the 6–9% figure

Reducing to 6–9% is ≈ 1.05–1.22 log units of attenuation. Three findings show the percentage form is not what the literature supports:

1. **The threshold is absolute, not relative.** Acuity is flat on the plateau and degrades only near $\sim 1\,\mathrm{cd/m^2}$ (8; 11) or the photopic/mesopic boundary. A fixed percentage maps to different absolute levels for different baselines.
2. **A $\sim 90\%$ cut on the plateau is minor.** At 0.06–0.067 logMAR/decade (15; 11), a 1.1-decade cut predicts only $\sim$0.06–0.08 logMAR (3–4 letters), within test–retest noise.
3. **Contrast sensitivity degrades first.** The binding constraint is CS/field (6) and the De Vries–Rose→Weber knee (~ 300 Td) (16), not high-contrast acuity.

The 6–9% value coincides with the photopic/mesopic edge *only* for a dim ($\sim 100\,\mathrm{cd/m^2}$) baseline; it has no independent citation as a percentage.

4 Conclusion and recommended threshold

No single validated percentage threshold exists in the literature. The closest defensible proxy is an *absolute* retinal-illuminance floor:

$$E_{\mathrm{ret}} \;=\; L_{\mathrm{scene}}\, A_{\mathrm{eff}} \;\gtrsim\; 300 \text{ Td} \quad \text{(Weber/CSF-saturation knee, 16)}, \tag{3}$$

or, in adaptation-luminance terms, keeping the eye above the ~ 3–$5\,\mathrm{cd/m^2}$ photopic floor (CIE 191:2010). For realistic daylight baselines this floor is met even for apertures well below 9% transmission. The simulation should therefore compute E_{ret} explicitly via Eq. (1), compare against the 300-Td knee, and apply the pupil-dilation and neural-adaptation corrections of Kim et al. (6) and Artal & Manzanera (2). A percentage 90% cut can cross into the mesopic regime and genuinely degrade acuity and contrast (5).

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